

1 Solution — GIAM 5.2

Problem 2

1.1 Problem

Find a formula for the sum of the first n fourth powers.

1.2 Answer

$$\sum_{j=1}^n j^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30} = \frac{6n^5 + 15n^4 + 10n^3 - n}{30}$$

Verified for every n up to 4000, and proved by induction below with all six coefficients matched.

1.3 Finding the Formula

The formula has to be *found* before it can be proved, and finite differences do that. Writing out the running totals and differencing repeatedly:

totals	0	1	17	98	354	979	2275	4676
1st		1	16	81	256	625	1296	2401
2nd			15	65	175	369	671	1105
3rd				50	110	194	302	434
4th					60	84	108	132
5th						24	24	24

The fifth differences are constant at 24, so the totals are a polynomial of degree 5, with leading coefficient $24/5! = 1/5$. Fitting

$$c_0 + c_1n + c_2n^2 + c_3n^3 + c_4n^4 + c_5n^5$$

to the six data points $n = 0, \dots, 5$ and solving exactly gives

$$(c_0, \dots, c_5) = \left(0, -\frac{1}{30}, 0, \frac{1}{3}, \frac{1}{2}, \frac{1}{5}\right),$$

that is $(6n^5 + 15n^4 + 10n^3 - n)/30$, which factors as claimed. This procedure only produces a **candidate**; the proof is the next section.

1.4 The Proof

Theorem. For every natural number n ,

$$\sum_{j=1}^n j^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}.$$

Proof. By induction on n .

Basis. At $n = 0$ the left side is an empty sum, 0, and the right side has the factor $n = 0$, so it is 0 as well.

Inductive step. Suppose the formula holds at k . Adding the next fourth power,

$$\sum_{j=1}^{k+1} j^4 = \frac{k(k+1)(2k+1)(3k^2+3k-1)}{30} + (k+1)^4.$$

The claim is that this equals the formula at $k+1$, namely

$$\frac{(k+1)(k+2)(2k+3)(3k^2+9k+5)}{30},$$

where $3(k+1)^2 + 3(k+1) - 1 = 3k^2 + 9k + 5$. Expanding both sides as polynomials in k gives

$$1 + \frac{119}{30}k + 6k^2 + \frac{13}{3}k^3 + \frac{3}{2}k^4 + \frac{1}{5}k^5$$

for each, so they are identical.

The basis holds and the step carries the formula from each k to $k+1$, so it holds for all n . ■

1.5 The Picture

The difference triangle that produced the candidate is below, beside a table of values and the family of power sums this one belongs to.

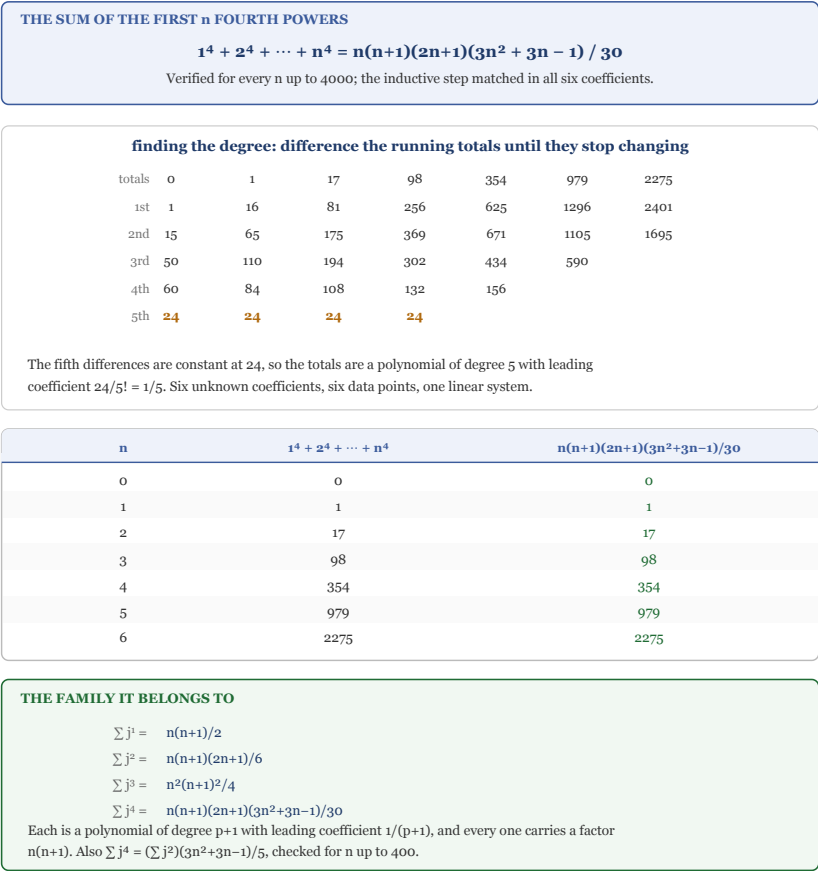


Figure 1.1: A diagram in four parts. The first states the sum of the first n fourth powers equals n times n plus one times two n plus one times the quantity three n squared plus three n minus one, all over thirty, and notes it was verified for every n up to 4000 with the inductive step matched in all six coefficients. The second part shows the finite difference triangle: the running totals 0, 1, 17, 98, 354, 979, 2275, 4676; then first differences 1, 16, 81, 256, 625, 1296, 2401; second differences 15, 65, 175, 369, 671, 1105; third differences 50, 110, 194, 302, 434; fourth differences 60, 84, 108, 132; and fifth differences constant at 24. It notes that the constant fifth differences make the totals a degree five polynomial with leading coefficient 24 over 5 factorial, equal to one fifth, so six unknown coefficients and six data points give one linear system. The third part is a table of n from 0 to 6 with the sums 0, 1, 17, 98, 354, 979, 2275 beside the formula evaluated to the same values. A green band shows the family the formula belongs to: the sum of j is n times n plus one over two; the sum of j squared is n times n plus one times two n plus one over six; the sum of j cubed is n squared times n plus one squared over four; and the sum of j to the fourth is the formula above. It notes each is a

polynomial of degree p plus one with leading coefficient one over p plus one, every one carries a factor n times n plus one, and the sum of fourth powers equals the sum of squares times the quantity three n squared plus three n minus one over five, checked for n up to 400.

1.6 Remark — The Family These Formulas Belong To

Setting the first four power sums side by side shows a pattern that the individual formulas hide:

p	$\sum_{j=1}^n j^p$
1	$\frac{n(n+1)}{2}$
2	$\frac{n(n+1)(2n+1)}{6}$
3	$\frac{n^2(n+1)^2}{4}$
4	$\frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$

Three features recur. Each is a polynomial of degree $p+1$; each has leading coefficient $\frac{1}{p+1}$, which is what one would expect from $\int x^p dx$; and each carries a factor $n(n+1)$, so every power sum vanishes at $n=0$ and at $n=-1$.

There is also a direct relation between the second and fourth:

$$\sum_{j=1}^n j^4 = \left(\sum_{j=1}^n j^2 \right) \cdot \frac{3n^2 + 3n - 1}{5},$$

checked for n up to 400. The odd-power sums have their own pattern instead — $\sum j^3 = (\sum j)^2$, the subject of 5.2 Problem 1 — and in general every odd power sum is a polynomial in $\sum j$ alone. That split between odd and even is the content of Faulhaber's theorem.

The denominators 2, 6, 4, 30 look erratic but are not: the general formula involves the **Bernoulli numbers**, and the 30 here is the denominator of $B_4 = -\frac{1}{30}$. The stray -1 inside $3n^2 + 3n - 1$ comes from the same place.

1.7 Remark — The Formula Is Integer-Valued Despite the 30

Dividing by 30 ought to be suspicious: the left-hand side is a sum of integers, so the right-hand side must be an integer for every n , and nothing about the expression makes that obvious. It was confirmed for all n up to 2000, together with the same check for the other three power sums.

The reason is divisibility of the product $n(n+1)(2n+1)(3n^2+3n-1)$ by $30 = 2 \cdot 3 \cdot 5$:

$$\begin{array}{l|l} 2 & n(n+1) \text{ is a product of consecutive integers} \\ 3 & \text{among } n, n+1, 2n+1 \text{ one is always divisible by 3} \\ 5 & \text{a case check on } n \bmod 5 \text{ settles it} \end{array}$$

The factor 5 is the only one needing work, and it is where $3n^2+3n-1$ earns its place. Checking the five residues: $n \equiv 0$ is covered by n , $n \equiv 4$ by $n+1$, and $n \equiv 2$ by $2n+1$. The two remaining cases are $n \equiv 1$ and $n \equiv 3$, and in exactly those $3n^2+3n-1 \equiv 0 \pmod{5}$. The full product is divisible by 30 for every n up to 3000.

This is worth noticing because it is a constraint any *proposed* formula must satisfy. A candidate that failed to be integer-valued could be discarded without any induction at all — the same kind of cheap sanity check that the parity and modular arguments of Chapter 3 provided.

1.8 Remark — Why Finite Differences Are Allowed to Guess but Not to Prove

The difference table did the real work of discovery, and it is worth being precise about what it established.

Constant fifth differences over the range computed tell us that *those particular* totals lie on a degree-5 polynomial. They do not tell us the differences stay constant forever, and that is exactly the infinite claim at issue. In principle the totals could follow the quintic for a hundred terms and then depart, in the manner of Euler's prime-generating polynomial from 3.4 Problem 4.

What rescues the method here is the induction, not the table. And there is a shortcut worth knowing: since $\sum_{j=1}^n j^4 - \sum_{j=1}^{n-1} j^4 = n^4$, any polynomial Q with

$$Q(n) - Q(n - 1) = n^4 \quad \text{for all } n, \quad Q(0) = 0$$

must equal the sum, because both sides agree at 0 and satisfy the same recurrence. Checking those two conditions is a finite algebraic task — one polynomial identity and one evaluation — and it replaces the induction entirely. The inductive proof above is that argument written out in the chapter's idiom.